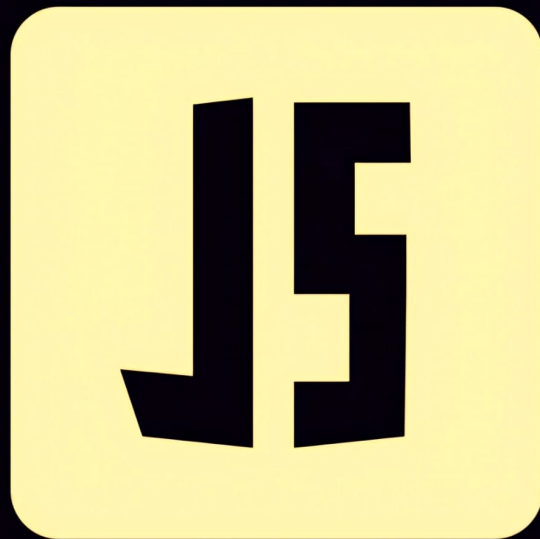


JavaScript

CheatSheet



1. Basic

Variables

```
let a = 10;           // block scoped
const b = 20;         // constant
var c = 30;           // function scoped (avoid using)
```

Data Types

```
let str = "Hello";    // string
let num = 123;         // number
let bool = true;       // boolean
let n = null;          // null
let u;                // undefined
let obj = { key: "val" };
let arr = [1, 2, 3];
```

Type Conversion

```
Number("123");        // 123
String(123);           // "123"
Boolean(0);            // false
```

2. Control Structures

if-else

```
if (a > b) {  
  console.log("A");  
} else {  
  console.log("B");  
}
```

switch

```
switch (day) {  
  case 1: console.log("Mon"); break;  
  default: console.log("Other");  
}
```

loops

```
for (let i = 0; i < 5; i++) {}  
while (condition) {}  
do {} while (condition);
```

3. Operators

```
+ - * / % **      // arithmetic
== === != !==    // equality
< <= > >=        // comparison
&& || !          // logical
= += -= *=       // assignment
```

4. Functions

```
function greet(name) {
  return `Hi, ${name}`;
}

// Arrow Function
const sum = (a, b) => a + b;

// Default Parameters
function greet(name = "Guest") {}

// Rest Parameters
function total(...nums) {
  return nums.reduce((a, b) => a + b);
}
```

5. Arrays

```
let fruits = ["apple", "banana"];

// Methods
fruits.push("mango");
fruits.pop();
fruits.shift();
fruits.unshift("kiwi");
fruits.includes("banana");
fruits.indexOf("apple");

// Iteration
fruits.forEach(f => console.log(f));
let newArr = fruits.map(f => f.toUpperCase());
let filtered = fruits.filter(f => f !== "banana");

// Spread
let all = [...fruits, "grape"];
```

6. Objects

```
let user = {
  name: "John",
  age: 25,
  greet() {
    return `Hi, ${this.name}`;
  }
};

console.log(user.name);
delete user.age;

// Destructuring
const { name } = user;
```

7. ES6+ Features

```
// Template Literals
let msg = `Hello, ${name}`;

// Destructuring
let [a, b] = [1, 2];

// Spread & Rest
let obj2 = { ...user, city: "NY" };

// Optional Chaining
let city = user?.address?.city;

// Nullish Coalescing
let val = input ?? "default";

// Short-circuiting
let isAdmin = user.isAdmin && true;
```

8. DOM Manipulation

```
document.getElementById("id");
document.querySelector(".class");

let el = document.createElement("p");
el.textContent = "Hello";
document.body.appendChild(el);

// Events
el.addEventListener("click", () => alert("Clicked"));
```

9. Async JS

Callback

```
setTimeout(() => console.log("Hi"), 1000);

// Promise
fetch("api/data")
  .then(res => res.json())
  .then(data => console.log(data))
  .catch(err => console.error(err));
```

Async/Await

```
async function getData() {
  try {
    let res = await fetch("api/data");
    let data = await res.json();
  } catch (err) {
    console.error(err);
  }
}
```

10. Advanced Topics

Closure

```
function outer() {  
  let count = 0;  
  return function inner() {  
    count++;  
    return count;  
  };  
}
```

Closure

```
console.log(a); // undefined  
var a = 10;
```

this keyword

```
const obj = {  
  name: "JS",  
  print: function () {  
    console.log(this.name);  
  }  
};
```

Prototypes

```
function Person(name) {  
  this.name = name;  
}  
Person.prototype.sayHi = function () {  
  return `Hi ${this.name}`;  
};
```

Classes

```
class Animal {  
  constructor(name) {  
    this.name = name;  
  }  
  speak() {  
    return `${this.name} makes noise.`;  
  }  
}
```

11. Useful Methods

```
// Array
arr.reduce((a, b) => a + b);
arr.find(e => e === 3);
arr.every(e => e > 0);
arr.some(e => e === 2);
```

```
// String
str.includes("H");
str.split(" ");
str.toUpperCase();
str.replace("H", "h");
```

```
// Number
parseInt("123");
parseFloat("12.3");
```

```
// Object
Object.keys(obj);
Object.values(obj);
Object.entries(obj);
```